# Effect of ethanol concentration on cell membrane permeability of beetroots

## Research question

To what extent does the alcohol concentration of 25ml of ethanol solution ( 20%, 40%, 60%, 80%, 100%) affect the cell membrane permeability of beetroot, measured through light absorption of leakage of pigment from beetroot cells?

## Introduction and Background

Where I'm from, the United States, there is a very severe overconsumption of alcoholic beverages that leads to millions of deaths. As a member of society, I have taken responsibility to research the negative effects that alcohol has on our body, specifically on the cell membranes. Membranes provide stability and protection to the cell. In addition, the membrane is semipermeable, allowing molecules to diffuse into and out of the cell. Beetroot cell membranes have a structure composed of a phospholipid layer with embedded proteins, which form a barrier that controls the passage of substances in and out of the cell. Because the beetroot cell membrane has a bilayer composition that requires continuous interactions among phospholipids and proteins, this layer can easily be disrupted by hydrophobic compounds such as ethanol (Savole, 2025). The disruption is caused by the reaction between ethanol, lipid tails, and denatured proteins in the beetroot cells (Feng et al., 2022), thereby affecting the permeability of the beetroot cell membranes and allowing beetroot pigment to pass through easily (Patra, 2006).

Betanin in beetroot cells is an aqueous, reddish pigment kept in vacuoles (*Betalain - an Overview | ScienceDirect Topics*, n.d.), capable of leakage into other liquids if the phospholipid bilayer is compromised. It can be detected using a colorimeter at a wavelength of 538nm (Rimadani Pratiwi et al., 2025). The influence of ethanol concentration on membrane permeability can be detected using a colorimeter based on the absorbance of the betanin. It may seem simple, but the experiment allows for repeatable outcomes.

Temperature is a well-studied factor affecting membrane permeability, but the impact of different ethanol concentrations isn't explored as much in school environments. Ethanol is important for things like disinfectants and food preservation. By seeing how the pigment leaks at differing ethanol concentrations, the effect of chemical agents on membranes in controlled conditions can be studied. To directly connect the differences in pigment leakage to the ethanol, the other variables are kept constant, such as time and test tube sizes.

It is relevant to investigate the influence that the concentration of ethanol has on the permeability of cell membranes, as it represents how certain chemical compounds can work to affect cell structures. Ethanol is used daily as a disinfectant and as a laboratory solvent, and through the investigation of how different ratios of ethanol affect beetroot cells, it is demonstrated how membranes respond to exposure to varying amounts of chemical substances. Furthermore, by investigating how ethanol affects membrane permeability, additional evidence can be gathered that membrane stability relies on the molecular composition of lipids and proteins to function effectively, which is relevant to fields such as medicine and food preservation.

## Hypothesis

The hypothesis is that as the membrane becomes more permeable with increasing ethanol concentration, more pigment leaks out and higher absorbance readings result (*Betalain - an Overview | ScienceDirect Topics*, n.d.). This is because higher ethanol concentrations are expected to cause more destabilization to the cell membrane as well as more denaturalization to proteins, allowing betanin to move into the solution outside. This should show a clear trend in data that shows the effect of ethanol on cells.

## Methodology

| Variable | Reasoning | Method |
| --- | --- | --- |

| | | |
|---|---|---|
| Independent variable: Ethanol concentration (20%, 40%, 60%, 80%, 100%) | The concentration of ethanol was varied to see how increasing amounts affect the permeability of beetroot cell membranes. By changing only, the ethanol concentration, any differences in betanin leakage can be linked directly to ethanol, and not to other factors. | The 100% ethanol was diluted down using water into the exact percentages that were needed for the experiment |
| Dependent variable: Absorbance of the solution at 538 nm using a colorimeter | The absorbance shows how much betanin has leaked from the beetroot cells. A higher absorbance indicates that the cell membranes are more permeable, which allows for quantitative comparison between different ethanol concentrations. | The absorbance of the solution was measured using the school colorimeter by collecting the remaining mixture of beetroots juice |

| | | |
|---|---|---|
| Controlled variable 1: Beetroot cylinder size and shape – same diameter and length | Keeping them the same allows the experiment to focus on ethanol concentration as the variable affecting permeability. | Cylinders were cut to a uniform size using a cork borer to ensure that surface area did not influence the amount of pigment leakage. |
| Controlled variable 2: Volume of ethanol solution – same for all samples | to ensure that concentration differences, and not differences in volume, were responsible for variations in pigment leakage. | Test tubes were filled with the same amount of ethanol using a pipette so that each beetroot cylinder was submerged in the same volume of solution. |
| Controlled variable 3: Temperature | Since temperature can also affect membrane permeability, the solutions were placed in water baths set to the same temperature. This ensures that changes in absorbance are caused by ethanol concentration and not by heat. | controlled using water baths at set temperature. Along with a cork to cover the test tubes to ensure that the temperature is kept constant inside test tube. |

| | | |
|---|---|---|
| Controlled variable 4: Colorimeter calibration | The colorimeter was calibrated before each set of measurements to guarantee that the absorbance readings were accurate and consistent, reducing systematic errors. | The lab made sure to give accurate colorimeters and helped calibrate it beforehand. |

**Method summary**

**1.** Prepare ethanol solutions of the desired concentrations and label them.

**2.** Cut beetroot into uniform cylinders using a cork borer and knife, and rinse gently to remove surface pigment.

**3.** Place two beetroot cylinders into each ethanol solution in a test tube.

**4.** Place the test tubes in a water bath set to the required temperature for 2 minutes to equilibrate.

**5.** Remove the beetroot cylinder and measure the absorbance of the solution at 538 nm using a calibrated colorimeter.

**6.** Repeat five times for each concentration using fresh beetroot cylinders and fresh ethanol solutions to improve reliability.

while controlling all other factors. By repeating the trials and carefully handling all samples, the experiment reduces random and systematic errors. It ensures that the measured absorbance accurately reflects the effect of ethanol on cell membrane permeability.

## Risk assessment

There weren't any concerns that had to do with ethics, as the experiment was done with beetroot, but there were environmental concerns in the alcohol solution for the experiment.

| Safety Hazard | Reasoning Of Concern | Safety Precaution |
|---|---|---|
| Knife | It has the possibility that a person could injure themselves from its sharp edges and cause infections | To prevent this one must wear protection gloves when using the knife to cut beetroot. |
| Ethanol exposure | Exposing Ethanol to skin or contact with eyes, respiratory system can lead to many problems like irritation. While also being highly flammable. | Ethanol was kept away from any fire source while also handled in safety glasses and gloves to prevent irritation from contact with skin and eyes. The lab technician also disposed this solution appropriately. |
| Glass (beakers, test tubes) | There is risk of beakers and test tubes shattering when falling onto the ground if placed on a slanted surface. Which could ultimately cut or injure anyone it contacts. | The glassware is kept away from the edge of tables to prevent accidental contact with it and handled with care. |

*Photos of the experiment setup*


Figure 1. Labeled Diagram of the setup of the experiment showing cut beetroots and test tubes in a rack.

## Data Collection

In this experiment, the absorbance of the solution surrounding each beetroot cylinder was measured using a calibrated colorimeter at 538 nm. For each ethanol concentration (20%, 40%, 60%, 80%, and 100%), five trials were conducted using fresh beetroot cylinders and fresh ethanol solutions. By repeating the measurements multiple times, it was possible to account for natural variation in the beetroot tissue, minor inconsistencies during preparation, and handling differences.

The absorbance readings were taken immediately after removing the beetroot cylinders from the ethanol solutions, ensuring that the measurements reflected the immediate effect of ethanol on membrane permeability rather than changes that might occur over longer periods.

The colorimeter was calibrated before each set of measurements to guarantee consistency and accuracy across all trials.

## Data

**Table 1** (Raw Data): Table depicting the absorbance of beetroot after heating through 20%, 40%, 60%, 80%, and 100% ethanol.

| | Number of Repeated trials done in same concentration of Ethanol | | | | | | |
|---|---|---|---|---|---|---|---|
| Concentration of Ethanol | Trial 1 | Trial 2 | Trial 3 | Trial 4 | Trial 5 | Standard DV | Mean |
| 20% | 0.00 | 0.00 | 0.00 | 0.00 | 0.01 | 0.00 | 0.00 |
| 40% | 0.03 | 0.02 | 0.04 | 0.08 | 0.05 | 0.02 | 0.04 |
| 60% | 0.28 | 0.12 | 0.24 | 0.28 | 0.19 | 0.07 | 0.22 |
| 80% | 0.33 | 0.38 | 0.40 | 0.41 | 0.33 | 0.04 | 0.37 |
| 100% | 0.51 | 0.32 | 0.45 | 0.50 | 0.42 | 0.08 | 0.44 |

The data show that as ethanol concentration increases, absorbance rises, indicating more pigment leakage and therefore increased membrane permeability. Calculating the mean absorbance for each concentration will help reduce random errors and provide clearer trends that can be used for analysis.

## Qualitative Observations

The alcohol solution formed a clear gradient from light pink to bright pink after exposure to the beetroot. The higher the alcohol concentration, the redder the solution formed.

During the 5-minute exposure, the beetroot shrank in size noticeably as the alcohol was exposed.

## Sample calculations for processed data

Average absorbance = absorbance value for all 5 trials / 5 ---- Example 40%: (0.03 + 0.02 + 0.04 + 0.08 + 0.05)/5 = 0.04

Standard Deviation: Enter the absorbance values for trials 1 – 5 into an Excel sheet, and type =STDV while dragging the trials 1- 5 to calculate the Standard deviation of trials 1 – 5.

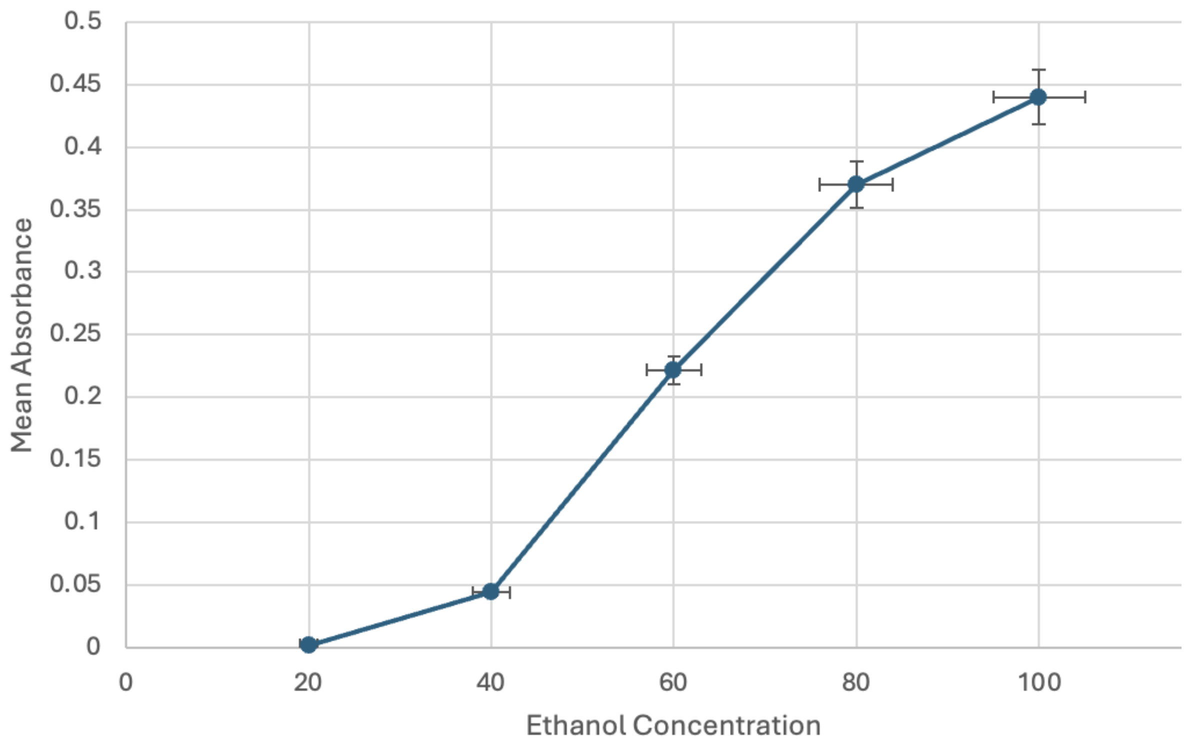
**Table 2**. Processed Mean Absorbance Calculated for each ethanol concentration (20%, 40%, 60%, 80%, 100%)

| Ethanol Concentration % | Mean Absorbance % |
|---|---|
| 20 | 0.00 |
| 40 | 0.04 |
| 60 | 0.22 |
| 80 | 0.37 |
| 100 | 0.44 |

After calculating the mean data from the observed data, as Table 1 shows, there is a very clear correlation between increasing concentration and mean absorbance

The uncertainty in the absorbance measurements was estimated based on the variability between trials and the precision of the colorimeter. Considering the fluctuations observed in the trials, along with the calibration accuracy of ±0.01, the overall uncertainty was approximately ±5% for each measurement. This was sufficient for observing trends in membrane permeability while acknowledging minor random errors.

Graph 1 showing the mean Absorbance compared to each Ethanol Concentration after 5 minutes.




Error bars represent the Standard Deviation

# Data analysis

As the ethanol concentration increases, so too does the average rate of absorption of the surrounding solution, indicating an increase in the leakage of betanin pigments from the beetroot cells. This shows that the permeability coefficient for the cell membranes is directly affected by ethanol concentration. For instance, at an ethanol concentration of 20%, little or no pigment leakage was observed, whereas at 100%, the rate of absorption of the surrounding solution was much higher.

The graph begins at 20% due to no trials being done with 0% ethanol, as there is ideally no change to the absorbance of beetroot when there is no alcohol present, so the effects of heating the beetroot at 0% ethanol were ignored.

The graph shows the standard deviation as error bars from 20% to 100%. Standard Deviation shows the variability of the data; smaller standard deviations mean that the data is more accurate and less scattered, while larger standard deviations mean that the data is less accurate and more spread out. In this experiment, the highest standard deviation was 0.08, which indicated that the data is very accurate, showing that the data in this experiment is reliable. A trend can be seen in the graph where the error bars are increasing with the rising ethanol percentages. This could be happening due to higher ethanol percentages giving beetroot membranes more fluidity for in and out movement, which results in a bigger range of absorbance data.

The membrane of beetroot cells is composed of a phospholipid-protein bilayer that's semipermeable in nature. Ethanol is a polar molecule and interacts with both the fatty tails as well as with the proteins of the membrane (Patra, 2006). Upon binding, ethanol disrupts the membrane structure and might even cause the denaturation of proteins. Such disruption renders the membrane more permeable. Consequently, water-soluble pigments like betanin leak out from the vacuoles into the surrounding solution. The data show that increased ethanol concentrations drive increased pigment leakage.

There are some variations from trial to trial, especially in the middle concentrations of ethanol, at 60% and 80%, where in 60% in trial 2, the value of 0.12 was lower than in the other trials. This can be accounted for by natural variations in the beetroot samples, as well as small differences in the sizes of the cylinders or their surface conditions. However, it still shows that with increased amounts of ethanol, there are increased amounts of absorbance.

The outcome of this experiment corresponds with the idea that for a membrane to remain stable, it must have a functional lipid and protein composition. Through this experiment, the role of chemical substances in destabilizing the membrane, causing it to leak, and releasing cellular material is shown. The application of this experiment can, therefore, be seen in bio-related contexts, such as where ethanol acts as a toxic material for living cells, as a disinfectant, a solvent, and a preservative.

In conclusion, the results are sufficient to confirm the positive direct relationship between ethanol concentration and cell membrane permeability. The consistent relationship between increased absorbance readings and the corresponding increases in ethanol percentages indicates the progressive disruption of the cell membrane and aligns with the known interaction between ethanol and the components found in the cell membrane. Despite the slight discrepancies shown in individual trials, this relationship still holds, supporting the assertion that increased ethanol concentrations result in increased pigment leakage.

# Evaluation

| Limitation | Explanation / Impact on Results | Improvement |
|---|---|---|
| Natural variation in beetroot samples | Beetroot is a natural material, so differences in cell density, pigment concentration, and membrane integrity between samples are unavoidable. This likely contributed to the variations seen between trials, especially at ethanol concentrations such as 60% and 80%. | Use beetroot cylinders taken from the same section of the beetroot and increase the number of trials to reduce the effect of natural variation. |
| Damage caused during cutting of beetroot cylinders | Cutting the beetroot damages surface cells, causing pigment to leak before the experiment begins. Although the cylinders were rinsed, this may not have been consistent for all samples, leading to slight differences in initial absorbance. | After cutting, soak the beetroot cylinders in distilled water for a longer period to remove excess surface pigment before placing them in ethanol solutions. |
| Colorimeter measurement uncertainty | Even after calibration, small fluctuations in absorbance readings can occur due to instrument sensitivity, cuvette positioning, or differences in solution clarity. This is more significant at low ethanol concentrations where absorbance values are very small. | Calibrate the colorimeter before each set of measurements and ensure cuvettes are positioned consistently. Increasing repeats would also reduce random error. |
| Temperature fluctuations during experiment | Although a water bath was used, slight temperature changes may have occurred when test tubes were transferred. Since temperature can affect membrane permeability, this may have influenced | Allow solutions more time to equilibrate in the water bath and monitor the temperature throughout the experiment to ensure consistency. |

| Limitation | Explanation / Impact on Results | Improvement |
|---|---|---|
| | pigment leakage alongside ethanol concentration. | |

# Conclusion

This experiment investigated the effect of varying levels of ethanol on the cell membranes of beetroot using the measure of betanin pigment leakage. The results show that a high concentration of ethanol results in a higher leak. At lower concentrations, such as 20%, very little pigment escapes, indicating that "the membranes remain largely unbroken." However, at higher levels, such as 80% and 100%, there is a significant increase in pigment leakage, which indicates serious damage to the membranes.

The net effect is that an increase in ethanol concentration increases the permeability of the membrane. Ethanol molecules bind with the phospholipid bilayer and proteins in the membrane and cause them to become disorganized and denature the proteins (Feng et al., 2021). This results in the weakening of the semipermeable membrane, and the water-soluble compound betanidin diffuses out of the vacuole and into the solution.

Some variation existed throughout the trials, especially regarding the medium levels of ethanol, which can be attributed to natural differences in beetroot samples and slight uncertainty in experimentation. Nonetheless, some similarity existed in that it was possible to indicate that a relationship exists between ethanol level and membrane permeability.

In conclusion, this experiment supports the hypothesis that certain substances, such as ethanol, can affect cell membranes in a concentration-dependent manner. Applications based on this process, such as disinfectants, solvents, and preservatives, can be observed in relation to ethanol.

## Extensions

An extension to this experiment can be to look at a wide range of different alcohols, such as methanol or propanol, instead of just looking at the concentration change of ethanol. Therefore, a research question could be: To what extent do different types of alcohol affect the cell membrane permeability of beetroot, measured through the absorption of light through a colorimeter? This can lead to further exploration of how drinking can impact our bodies, according to the type of alcohol. By investigating how different alcohols damage cells through the effects of leakage in beetroot can gain clearer insight on if one should continue drinking and the effects that alcohol could bring.

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